Introduction to scientific research and writing a research proposal

A SHORT INTRODUCTION

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Goals of this lecture



- What is research?
- How to make strong research questions?
- How to write a research proposal?

What is research?

- Research refers to a **search for knowledge**
- Research means *a* **scientific and systematic search** for pertinent information on a specific topic
- The purpose of research is to **discover answers** to questions through scientific procedures



Why bother being scientific?

- We are assaulted every day by a mass of **opinions** and **alternative facts** regarding how the world works
- The challenge is to know which ones of these we should believe
- In approaching a question scientifically
  - we attempt to take superstition, prejudice and guess-work out of this problem and
  - provide systematic grounds for why our conclusions should be believed



Being scientific

- This entails being:
  - Systematic - applying consistent methods
  - Empirical - in the sense of 'drawing conclusions based on evidence'
  - Rigorous - being thorough and precise
  - Sceptical - being aware of possibility of disconfirmation
  - Ethical
  - …and probably other things too, but it's a start!
- To count as 'scientific', conclusions need to be
  - compelling and
  - communicable



Research and Development

- **Research**: discover new knowledge (new regularities)
  - Basic research (no specific application in mind)
  - Applied research (ideas into operational form)
- **Development**: systematic use of the existing knowledge
  - Research and development are closely related
  - In research a prototype is often developed

How to make a strong research question

Why making a research question?

- When **starting** a new research project, it is important to develop a solid research question.
- Developing the right research question is **a critical first step** in the research process as it will guide all research activity.

- https://research.com/research/how-to-write-a-research-question



Importance of a research question



Focus - The primary importance of framing the research question is that it **narrows down** a broad topic of interest into a specific area of study



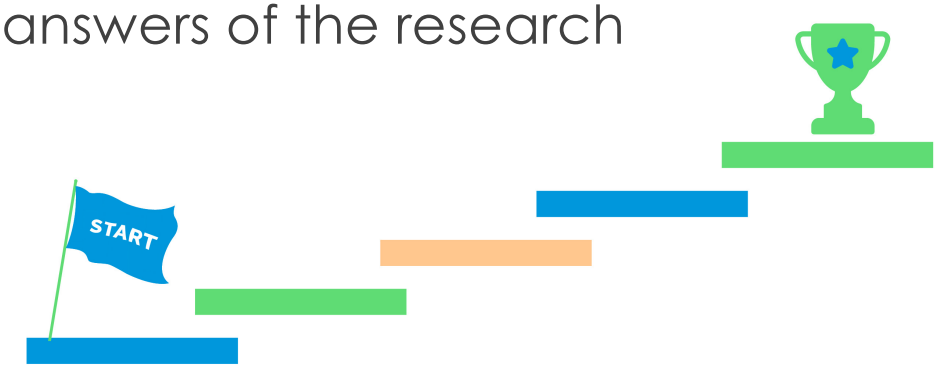
Guideline - Research questions, along with hypotheses, also serve as a **guiding framework for research**. These questions also specifically reveal the boundaries of the study, setting its limits, and ensuring cohesion.

What is a Research Question?

- A research question is a question that a study or research project *aims to answer*.
- This question addresses an issue or a problem, which, through analysis and interpretation of data, is answered in the study's conclusion.
- While many research projects will focus on a single research question, larger studies often use more than one research question.

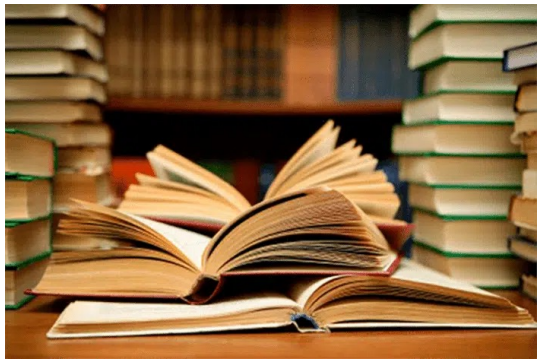
Steps to take

1. Preliminary review of related literature
2. Narrow down your topic and determine potential research questions.
3. Evaluate the soundness of your research question.
4. Identify metrics to evaluate the answers of the research question



1. Preliminary review of related literature

- This initial stage of research accomplishes two goals.
  - allows you to discover issues that are currently being discussed by fellow researchers.
  - allows you to spot existing gaps or limitations in existing knowledge of your topic.
- In this process, it is vital to ask questions such as:
  - What has been done before, and more recently? How were these studies conducted? What hypotheses were tested?



2. Narrow down your topic

- **Gap-spotting** - focus on gaps in existing knowledge or recent literature.
  - this method involves constructing research questions out of identified limitations in literature and overlooked areas of study.
- **Extend** - Choose research questions that extend or complement the findings of existing literature.
  - For example consider the existing knowledge in an other domain and context, or with different assumptions
  - Patterns and trends in practice may also provide new insights and potential ideas for research questions.



3. Evaluate soundness

A research questions should be:

- **Focused** on a single problem or issue
- **Researchable** using primary and/or secondary sources
- **Feasible** to answer within the timeframe and practical constraints
- **Specific** enough to answer thoroughly
- **Complex** enough to develop the answer over the space of a paper or thesis
- **Relevant** to your field of study and/or society more broadly
- **Original**, so interesting and not done before

Characteristics of a strong research question



4. Identify metrics

- Identify metrics to evaluate the answers of the research question
- So ask yourself:
  - What data should I gather to get an answer to the question?
  - How could I represent that, e.g. in graphs or tables?
  - Does this indeed show the answer to the question?
- A research question is never to be answered by yes or no

Example 1

- How does social media affect people's behavior?
-



Example 1

- **Bad**: How does social media affect people's behavior?
- **Good**: What effect does the daily use of YouTube have on the attention span of children aged under 16?

- The first research question is considered bad because of the *vagueness* of "social media" as a concept and the question's *lack of specificity*.
- A good research question should be *specific* and *focused*, and its answer should be discovered through *data collection and analysis*.



Example 2

- **Bad**: Has there been an increase in childhood obesity in the Netherlands in the past 10 years?
-



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- **Good**: How have school intervention programs and parental education levels affected the rate of childhood obesity among undergraduate students?



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- The first research question is not ideal because it's *too simple*, and it's easily answerable by a "yes" or "no."
- The research question is more *complicated*; to answer it, the researcher must *collect data*, perform in-depth *data analysis*, and form an *argument* that leads to further discussion.

Example 3

- **Bad**: Why is there a housing crisis in the Netherlands?
-



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- Starting with "why" often implies that your question is *not focused* enough: there are too many possible answers.
- By targeting just one specific aspect of the problem, the second question offers a *clear path* for research.



Example 4

- **Bad**: How can drunk driving be prevented?
-



More examples: https://www.scribbr.com/research-process/research-question-examples/

Example 4

- **Bad**: How can drunk driving be prevented?
- **Good**: What effect do different legal approaches have on the number of people who drive after drinking in European countries?



- The first question asks for a ready-made solution, and is *not focused* or researchable.
- The second question is a *clearer* comparative question, but note that it may not be practically *feasible*. For a smaller research project or thesis, it could be narrowed down further to focus on the effectiveness of drunk driving laws in just one or two countries.

More examples: https://www.scribbr.com/research-process/research-question-examples/

Key points to remember

- A research question pinpoints exactly what you want to find out in your work.
- A good research question is essential to guide your research.
- Make the research question as specific and concise as possible to ensure clarity.
- Identify the metrics upfront, this will guide you towards the answers that you need

Writing a research proposal

Why a research proposal?

- Convince others the project you have designed is important, worth the effort.
- Convince others that you have the ability to carry out the research design and report the findings.
- It is like a blue print of a building plan before the construction starts



"You are completely free to carry out whatever research you want, so long as you come to these conclusions."

Research proposal

- Identify a compelling research question
  - And focussed!
  - Should be feasible to finish in the time allocated
  - Guideline: aim at research for a period of (only) 5 weeks
- Elaborate and write the proposal in a document
  - Only 3 pages.



What should the research be about?

- Possible topics
  - Basically anything in the domain of the tracks of the module
  - The track chair could give some hints
  - Technologies, protocols, algorithms, applications, ..
- New solutions, new techniques, new models, etc.
  - To solve a problem in the domain
  - It should bring new contributions
    - So it is not about developing new systems
    - Although a prototype or proof of concept is extremely useful!

What makes a good research proposal?



A well-prepared application should require minimal effort on the part of the reviewer.



Proposals must demonstrate high scientific quality.



The application should be feasible in the envisioned time-line and available resources

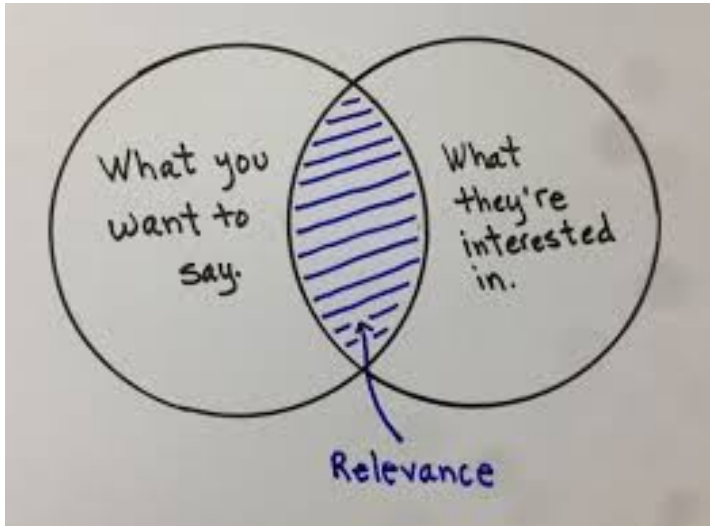
What should a research proposal include?

- 1. Introduction
  - Objective and goals
  - Research question
  - Justification
  - Background /Review of literature
- 2. Methodology and approach
- 3. Work programme
  - Tasks, results
  - Time frame and work schedule/Gantt chart



1. Introduction

- Introduce the problem
- Provide the context
- Indicate its importance and relevance
- Review the state of the art
  - Indicate related work and use appropriate references
  - Identify missing research



Objectives and justification

- This is a very important and pivotal section
  - everything else in the study is centered around it
- The objective of the proposed study should be
  - stated very clearly
  - specific, achievable and measurable
- Too many objectives to be avoided



Research question

- A research question pinpoints exactly what you want to find out in your work.
- A good research question is essential to guide your research.
- Make the research question as specific and concise as possible to ensure clarity.



2. Methodology and approach

- Research methodology is a way to systematically solve the research problem
- Identify methods, techniques and metrics for evaluation
- Identify whether an ethical approval is needed!



3. Work programme

- A study design is a specific plan or protocol for conducting the study
  - allows the investigator to translate the conceptual hypothesis into an operational one.
  - The study design to be used should be appropriate for achieving the objective of the study
- Draft a research plan
  - Separate tasks
  - Identify intermediate results and milestones
  - Provide a suitable timeline for the sequence of tasks
  - Use flow charts and diagrams
  - And aim to finish well in time: be realistic, and don't be too ambitious.



Planning types



Global Timeline

For you and your supervisor to track at what stage you are. With clear and larger goals.

You may update this every 3-4 weeks or as requested by your supervisor.

Use a **Gantt chart** for that with separate tasks and deadlines



Weekly Timeline – This is for YOU!

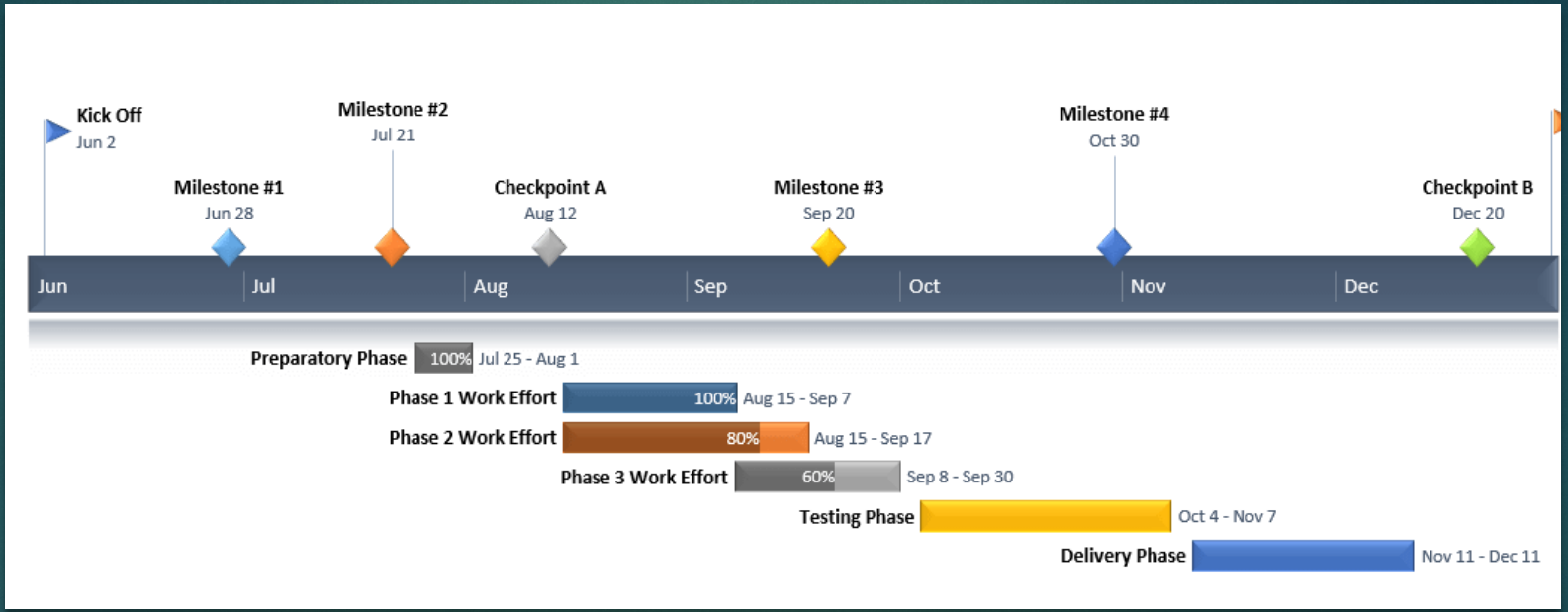
Plan daily activities, with expected results

This should be updated weekly, and checked for allignment of Global timeline

Gantt chart

- A Gantt chart is a visual description of a project's timeline. The chart shows the start and end dates of a project's components, such as resources and planning.
- It typically includes two sections:
  - a list of tasks,
  - a timeline with schedule bars that visualize work.
- Gantt bars show start and end dates, as well as dependencies, scheduling and deadlines.
- Also include the expected results
  - deliverables, demo, presentation, drafts, etc.





https://www.officetimeline.com/gantt-chart/how-to-make/powerpoint

Planning via a Gantt chart

- Think through all the tasks involved and divide them into manageable components.
- A Gantt chart will show the tasks in a sequential order and display task dependencies
  - i.e. how the outcome of one task relates to another (milestones).
- Determine and allocate your resources.
- Anticipate the risks and problems you may encounter and create a contingency plan for potential problems.

Planning tips



1. Clear problem statement and research questions

Specific and concrete with attainable goals

Identify the metric for evaluation upfront that will answer the ressearch question



2. Set pragmatic goals and make a contingency plan

Identify risks and evaluate consequences

Consider all the external variables that are beyond your control



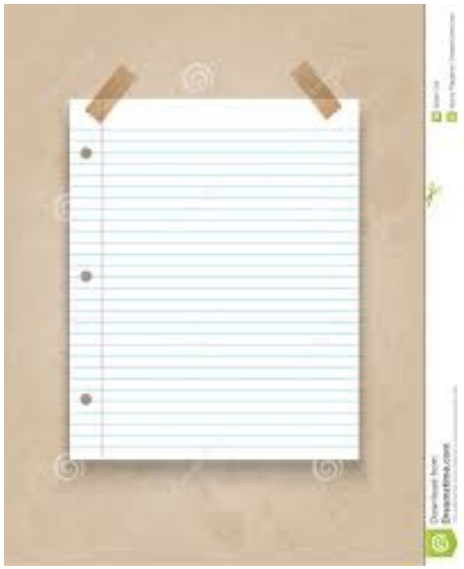
3. Set realistic timelines

Consider the *limited availability of time and resources*, and the possibility that you may need to *revisit and refine* your problem statement and objectives

Prioratize - Identify the **minimum results** needed, and *be nice to have*

Style

- Although it is the content that matters, good presentation is often crucial to making your proposal accessible to reviewers and keeping their interest.
  - Use diagrams and tables to add clarity;
  - Bullet points and sections can break up text;
  - Keep to page, word and font size restrictions; and
  - Activate the spell checker while writing.
- Title and abstract are the essential part of a proposal
  - It needs to be clear and concise!



What are reviewers looking for ?

- High scientific quality;
- Proposals that meet the funder's priorities or fill a knowledge gap;
- Novelty and timeliness;
- Value for money;
- A clear and well thought out approach; and
- An interesting idea – catch their attention!

Final remarks

- Research proposal requires real work
  - Review literature
  - Identify challenges and research questions
  - Focus!
    - better to have a well identified and feasible research goal,
    - than a generic and too ambitious research goal

Conclusions

- Writing a research proposal requires
  - Knowledge about topic
  - Identification of challenges or missing aspects
  - Identification of goals and objectives
  - Formulate research questions
  - Planning towards achieving those goals
  - And make them measurable

Useful further reading

- https://www.scribbr.com/category/research-process/
- And with a nice quiz at https://www.scribbr.com/research-process/research-questions/
- https://research.com/research/how-to-write-a-research-question